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1 Basic Counting and Sum Algorithms for Streaming Data

[3]:
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# Basic Counting Problem for Stream Data

class CountStream:
    def __init__(self):
        self.count = 0

    def update(self, _item):
        # Increment count for each stream element
        self.count += 1

    def value(self):
        # Return current total count
        return self.count

# Example:
counter = CountStream()
for element in [10, 7, 3, 5, 2]:
    counter.update(element)
print("Total elements counted:", counter.value())
```

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Total elements counted: 5
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[4]:
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#  Sum of Elements in a Data Stream (Integers)

class SumStream:
    def __init__(self):
        self.total = 0

    def update(self, x):
        # Add each stream element to sum
        self.total += x

    def value(self):
        # Return current sum
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        return self.total

# Example:
sumi = SumStream()
for element in [10, 7, 3, 5, 2]:
    sumi.update(element)
print("Sum of elements (integers):", sumi.value())
```

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Sum of elements (integers): 27
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[6]:
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#  Sum of Elements Using Kahan Summation (Floats)

class SumStreamFloat:
    def __init__(self):
        self.total = 0.0
        self.c = 0.0

    def update(self, x):
        # Kahan summation improves accuracy with floating point data
        y = x - self.c
        t = self.total + y
        self.c = (t - self.total) - y
        self.total = t

    def value(self):
        return self.total

# Example:
sumf = SumStreamFloat()
for element in [1.1, 2.2, 3.3, 4.4, 5.5]:
    sumf.update(element)
print("Sum of elements (floats, Kahan):", sumf.value())
```

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Sum of elements (floats, Kahan): 16.5
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